



Social Distancing Alarm Cap

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Here is an Arduino based social distancing alarm cap meant for Covid-19 pandemic or similar disease outbreak. Maintaining the right gap for social or physical distancing is not always feasible. When we are outside, we forget to maintain the distance as soon as we get involved in our daily work.

This novel circuit alerts you if you come too close to someone either from the front, back, right, or left

direction. A cap fitted with four ultrasonic distance sensors connected to Arduino Uno senses the minimum safe distance and alerts you through a buzzer. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the social distancing alarm cap is shown in Fig. 2. It is built around Arduino Uno, four HC-SR04 ultrasonic sensors, and a few other components.

HC-SR04 ultrasonic sensor utilises sonar to determine distance from an object (or humans as in this case). It offers excellent non-contact range detection with high accuracy and stable readings in an easy-to-use package. Its range is from 2cm to 400cm with an accuracy of 5mm. The module consists of an ultrasonic transmitter, receiver, and

control circuit. HC-SR04 ultrasonic sensor is shown in Fig. 3. It has the following four pins, namely:

1. VCC: +5V DC
2. Trig: Trigger (input)
3. Echo: Output
4. GND: Ground

It can be easily interfaced with Arduino Uno.

The working of HC-SR04 is simple. When the sensor is triggered (a short 10µs pulse to the trigger input pin 2 from Arduino Uno board), the transmitter sends a high-frequency sound signal at 40kHz. The transmitted signal reflects back from the nearby object and is picked up by the receiver.

The output of HC-SR04 (echo-pin 3) is a pulse where the width of the pulse is proportional to the distance of the object. The width of the pulse can be calculated by the `pulseIn()` function in the code. Arduino program calculates the distance of the objects in all its four directions.

Here, four ultrasonic sensors are placed in four directions and connected to Arduino Uno. Each sensor collects the distance of the object in all four directions and compares it with the distance safety value defined in the program. When the user comes too close to another person in any of the directions, an alarm sound is produced.

Pin connections of Arduino with components are shown in the Table.

Software

The circuit operation is performed through the software program loaded into the internal memory of Arduino Uno. The program is simple and easy to understand. Comments are given at the end of each command line.

The code starts with defining the pin numbers. Trigger and echo pins are needed for each sensor. So, a total of eight pins are needed to connect to the Arduino board. The buzzer is



Fig. 1: Author's prototype

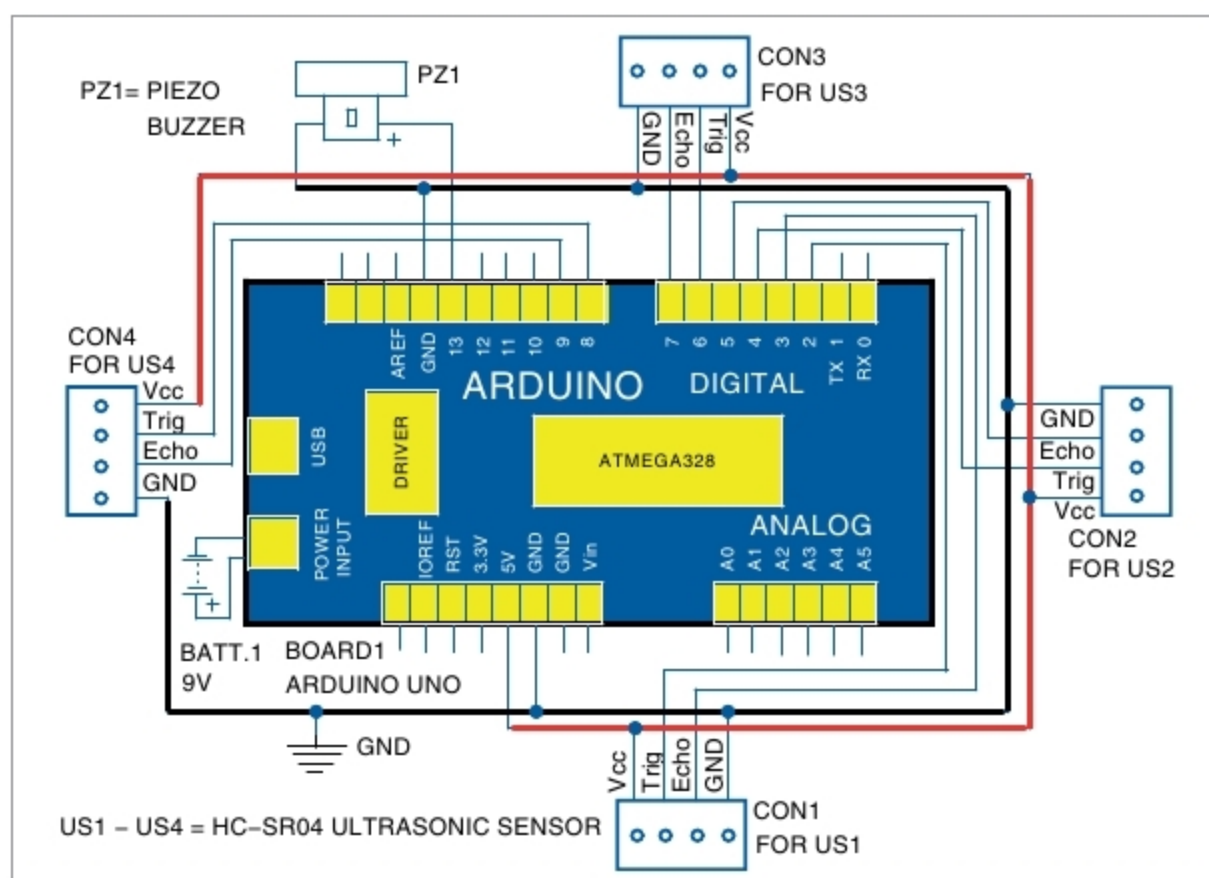


Fig. 2: Circuit diagram of social distancing alarm